ATTACHMENT A –
TECHNICAL SPECIFICATIONS

1. **SIGNALS (Requirements)**
   The technical characteristics of the signal generated by the system must meet the following standards:

   - Standard ITU-R BS 450 (system frequency pilot);
   - Standard ITU-R BS 643 (RDS).

2. **Description of the ANTENNA SYSTEM**
   The antenna system used for the irradiation of the broadcast signals of the issuing AFN is currently in use. It is expected the use of a radiant system in VHF band, consisting of:

   - collinear antenna composed of n. 4 dipoles in pol. V;
   - electrical center at 128 m from the base of the tower;
   - vertical encumbrance of 8,88 m.

3. **Appendices to Attachment A:**

- The geometry of the antenna and the characteristics of the feeders (Appendix A1);
- The technical card of the broadcasting station, working on the frequency 97.300 MHz , with the indication of characteristics of radiation in the different directions of propagation (Appendix A2).
- Coverage Maps (Appendix A3)

APPENDIX A1 GEOMETRY OF THE ANTENNA AND THE CHARACTERISTICS OF THE FEEDERS

TX station: AFN Naples | Locality: Camaldoli 97.3 MHz

General data of antenna System

TX station	AFN Naples
Locality	Camaldoli 97.3 MHz
Description	
Status	Attivo
System of coordinates	WGS84
Longitude	14°12'02.065"
Latitude	40°51'36.434"
Ground level a.s.l. (m)	402.0
Antenna system height (m)	128.0
Transmitter power(Watt)	5000.000
Carrier wave frequency (MHz)	97.300
Antenna system central frequency (MHz)	97.000
Antenna base diagrams type 1	KATHREIN-775130 LB FM dipole Flg. 1,5_8
Polarization (H/V/C/X)	C
Transmitting cable attenuation (dB)	0.5
Additional attenuations(dB)	0.5
Base diagrams sectors (T = All, F = Front)	T
Velocity factor of cables to Antennas (0÷1)	0.93
Coordinate System(Cartesian, Polar, Offset)	C
Mast side / diameter(cm)	0.0
Mast cross section (T/Q/C)	Q
Structure rotation w.r.t. North (°)	0.0
Mast rotation w.r.t. North (°)	0.0

Information about antennas used in the System

	Antenna type 1
Manufacturer	KATHREIN
Antenna model	775130 LB FM dipole
Band start(MHz)	87.5
Band stop(MHz)	108
diagrams Frequency(MHz)	98
Polariz (H/V/C/X)	V
Vertical dist (cm)	280
Height (cm)	140
Width (cm)	8
Thickness (cm)	83
Weight (Kg)	13
Maximum power (KW)	5
Gain (dBd)	2.4
North E.C. (cm)	0
East E.C. (cm)	0
Return loss (dB)	0
R.C.Phase (°)	0

TX station: AFN Naples

Frequency: 97.30 MHz

Gain solid integration : enabled

Locality: Camaldoli 97.3 MHz

Antennas arrays data

Note: calculation of single antennas arrays data (without taking into account mutual effects)

A. Antennas array azimuth (°/N)	120
B. Number of antennas	4
C. Nominal power supply (W)	5000.00
D. Losses (addit. + cables) (dB)	1.0
E. Effective power supply (W)	3971.64
F. Theor. maximum gain (dBd)	8.53
G. Distribution losses (dB)	0.00
H. Nominal max gain F - G (dBd)	8.53
I. Compensation losses (dB)	0.07
J. Effec. max gain H - I (dBd)	8.46
K. Effec. max gain (times)	7.02
L. Effec. max power E * K (KW)	27.8752
M. Max power depr. angle (°)	3.8
N. Max power az. angle (°)	100

Diagram in dBK calculated at horizon

Az. (°/N)	dBK	Az. (°/N)	dBK	Az. (°/N)	dBK	Az. (°/N)	dBK
0	8.1	90	13.7	180	13.0	270	6.9
10	8.9	100	13.8	190	12.5	280	6.9
20	9.9	110	13.8	200	11.8	290	6.9
30	11.0	120	13.8	210	11.0	300	6.9
40	11.8	130	13.8	220	9.9	310	6.9
50	12.5	140	13.8	230	8.9	320	6.9
60	13.0	150	13.7	240	8.1	330	6.9
70	13.5	160	13.6	250	7.6	340	7.1
80	13.6	170	13.5	260	7.1	350	7.6

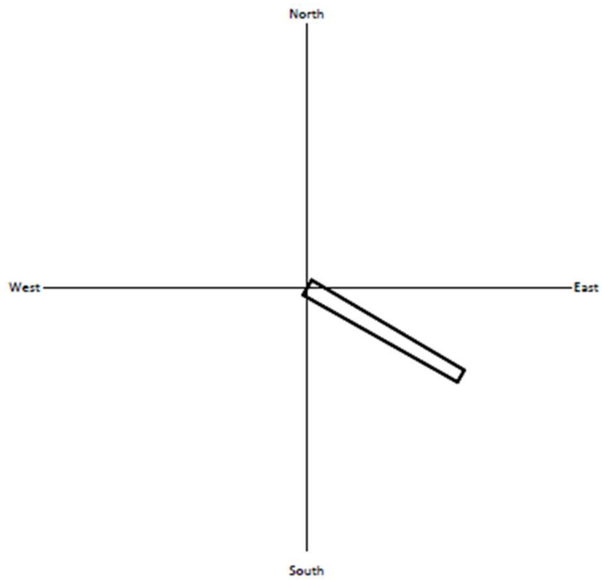
Diagram in dBK calculated at horizon
(without -20dB\'s lower limit vs maximum power)

Az. (°/N)	dBK	Az. (°/N)	dBK	Az. (°/N)	dBK	Az. (°/N)	dBK
0	8.1	90	13.7	180	13.0	270	6.9
10	8.9	100	13.8	190	12.5	280	6.9
20	9.9	110	13.8	200	11.8	290	6.9
30	11.0	120	13.8	210	11.0	300	6.9
40	11.8	130	13.8	220	9.9	310	6.9
50	12.5	140	13.8	230	8.9	320	6.9
60	13.0	150	13.7	240	8.1	330	6.9
70	13.5	160	13.6	250	7.6	340	7.1
80	13.6	170	13.5	260	7.1	350	7.6

Geometr. and electrical data of antenna System

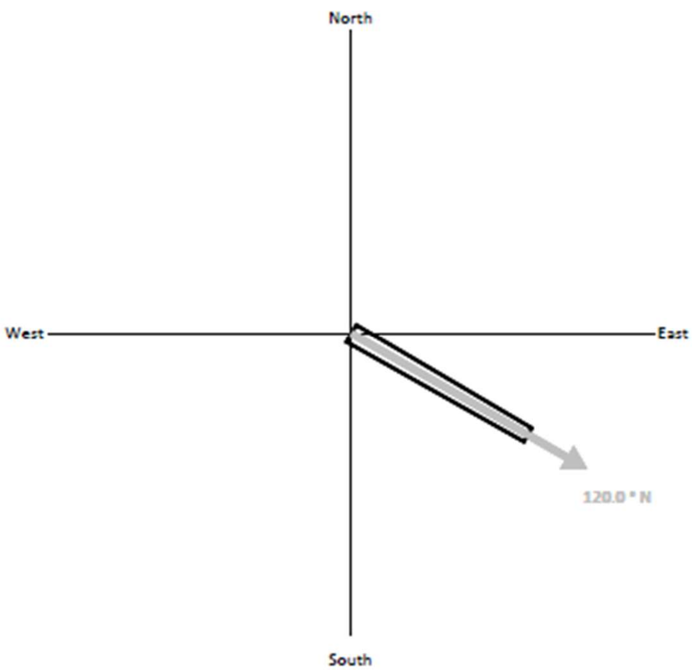
	Power (%)	*Tilt (°)*	*Az. (°/N)*	*Group Phase(°)*	*Phase (°)*	*V dist. (m)*	*E.C. X (cm)*	*N.C. Y (cm)*	*Rot. (1÷4)*	*Type (1÷2)*	*L cables (cm)*	*Car. phase(°)*
1	25.0000	0	120	0	+30.4	3.75	0.0	0.0	1	1	-24.3	30.5
2	25.0000	0	120	0	+10.1	1.25	0.0	0.0	1	1	-8.1	10.2
3	25.0000											-10.2
4	25.0000											-30.5

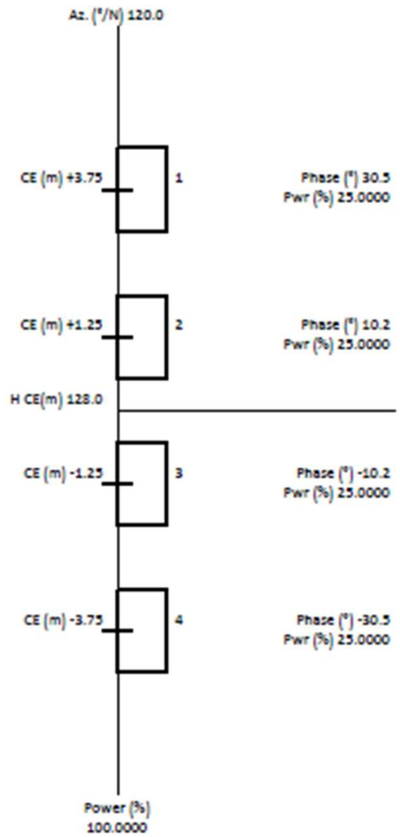
Plan of antenna system



Side of antenna system

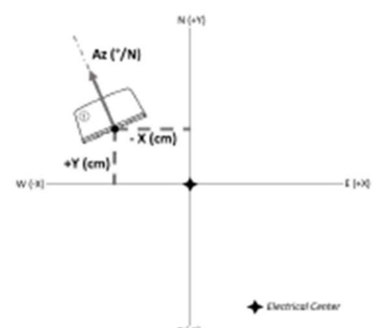
Array Details 1/1 - 120.0 °/N



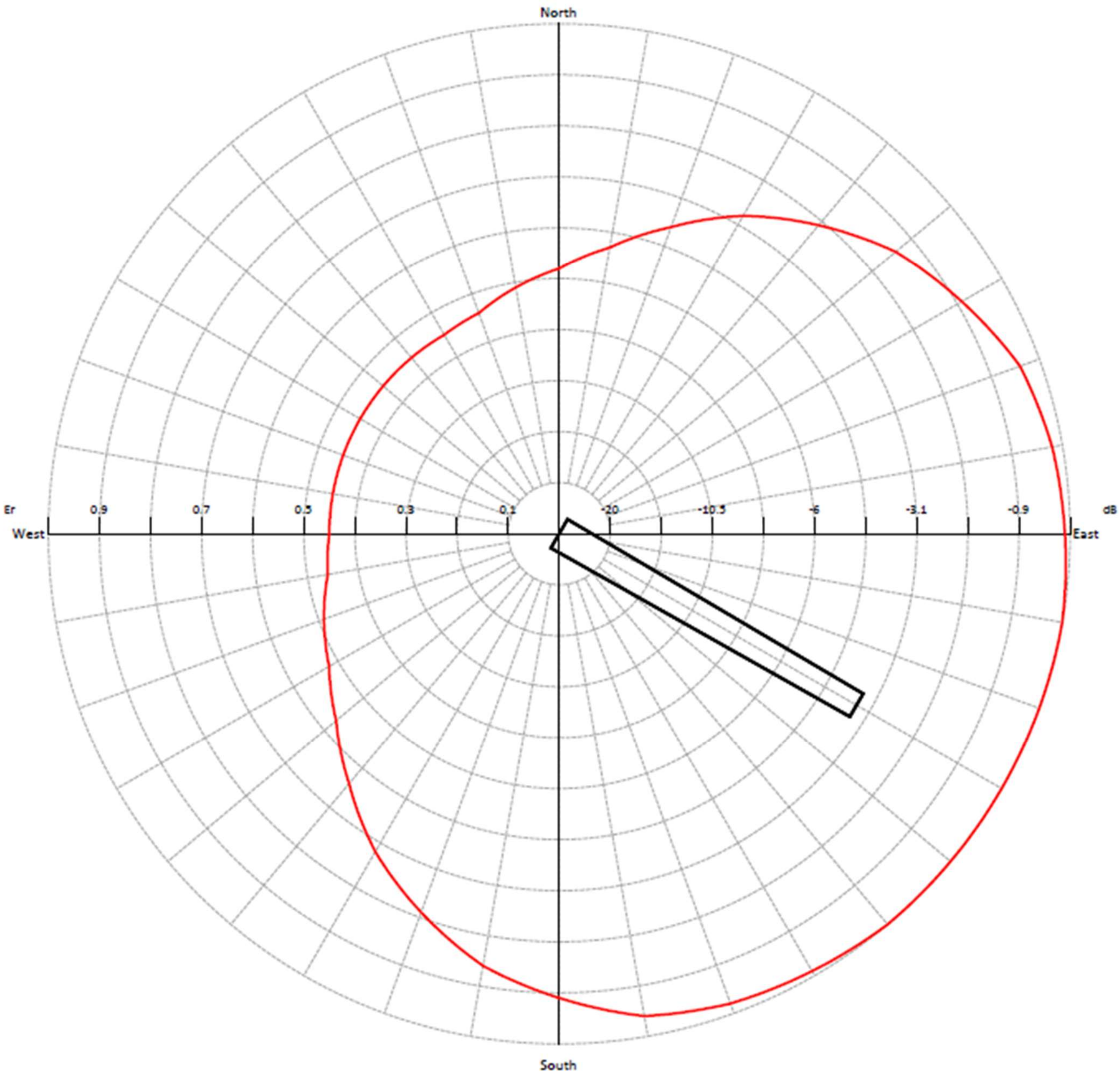


Geometr. and electrical data of Array 1/1 - 120.0 °/N

	Power (%)	*Tilt (°)*	*Az. (°/N)*	*Group Phase(°)*	*Phase (°)*	*V dist. (m)*	*E.C. X (cm)*	*N.C. Y (cm)*	*Rot. (1÷4)*	*Type (1÷2)*	*Car. phase(°)*
1	25.0000	0	120	0	+30.4	3.75	0.0	0.0	1	1	30.5
2	25.0000	0	120	0	+10.1	1.25	0.0	0.0	1	1	10.2
3	25.0000	0	120	0	-10.1	-1.25	0.0	0.0	1	1	-10.2
4	25.0000	0	120	0	-30.4	-3.75	0.0	0.0	1	1	-30.5



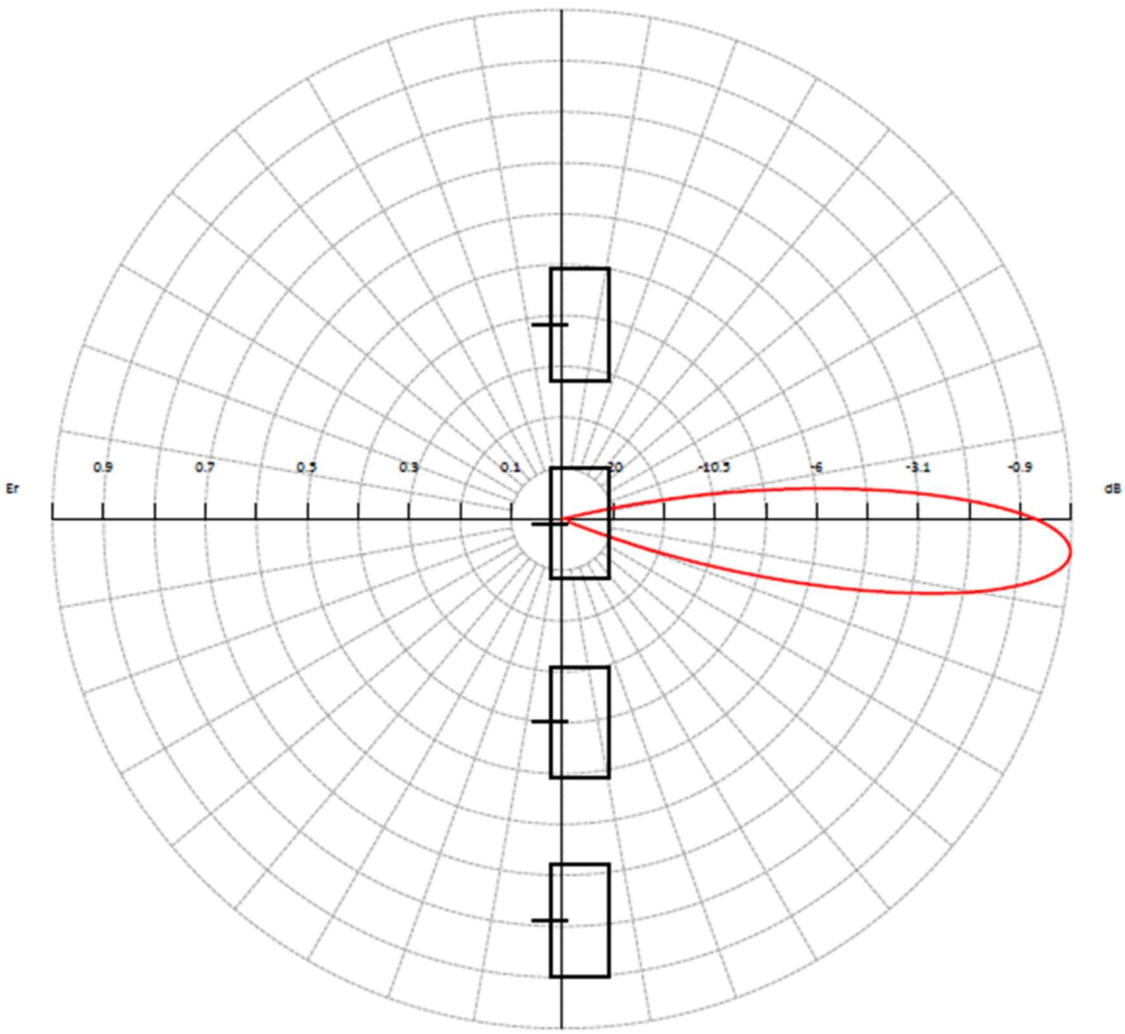
Horizontal diagram at 0.0° depres. (Total Antenna)



0.0° depres. (Total Antenna), Gain (dBd): 7.83

ERP T.Max(KW): 30.352 ERP E.Max(KW): 24.109

Vertical diagram at an azimuth of 120.0° degrees



120.0° Az. (Total Antenna), Gain (dBd): 8.46

ERP T.Max(KW): 35.093 ERP E.Max(KW): 27.875

APPENDIX A2

TECHNICAL CARD OF THE BROADCASTING STATION, WORKING ON THE FREQUENCY 97.300 MHZ , WITH THE INDICATION OF CHARACTERISTICS OF RADIATION IN THE DIFFERENT DIRECTIONS OF PROPAGATION

SCHEDA B - IMPIANTO PRIVATO RADIOFONICO [X] TV [ ]

ATTENZIONE Se l'impianto è di solo collegamento si deve rispondere soltanto ai punti 36,37,38,39,41 e dai punti dal 58 al 68.Per la messa in onda si risponde soltanto ai punti 36,37,38,39,40,41.Per i ripetitori di programmi esteri o nazionali non va compilata ovviamente la scheda di messa in onda, ma va indicato al punto 64 per il solo primo impianto della della catena, la stazione straniera o rai ricevuta, gli impianti successivi vanno trattati normalmente.

Radio: AFN AMERICAN FORCES NETWORK 36
Denominazione Emittente

37 | CAMALDOLI 37
N° Impianto

CAMALDOLI 38
Indirizzo

NAPOLI 39
Centro Abitato

Lombardi Ugo
American Forces Network
AFN Naples The Eagle
prot. 900552

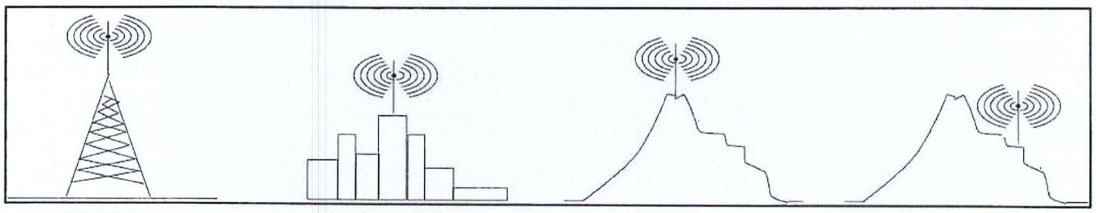
NA 40
Provincia

NAPOLI 41
Comune

14 | 11 | 59
Longitudine

40 | 51 | 34 43
Latitudine

402 44
Quota s l m



45	46	47	X 48
Traliccio	Palazzo	Vetta	Dorso

97,3 (50) Freq MHz	(51) Portante Audio (per TV)	(52) Canale TV	(53) Posiz. Offset TV	(54) Tipo di Offset
5 (55) Pot C KW	NAUTEL (56) Costr Apparato	2009 (57) Anno Costruzione		

35 (58) Riceve il segnale da	(59) o da 1	(60) o da 2	
P (61) Mediante	(62) Mediante	(63) Mediante	Eutelsat 9B (64) Riceve il seg da
406 (65) Freq MHz	(66) Freq MHz	(567) Freq MHz	11804 (68) Freq MHz

NA | CE | | | | | | | | | 69
Sigle provincie interessate dal servizio di radiodiffusione

P 70
Tipo di servizio

(71) Località 1 esclusa deliberatamente dal servizio	(72) Pv 11	(73) Metodo usato
(74) Località 2 esclusa deliberatamente dal servizio	(75) Pv 12	(76) Metodo usato
(77) Località 3 esclusa deliberatamente dal servizio	(78) Pv 13	(79) Metodo usato

SCHEDA C - IMPIANTO PRIVATO RADIOFONICO [X] TV [ ]

ATTENZIONE Se l'impianto è di solo collegamento si deve rispondere soltanto ai punti 36,37,38,39,41 e dai punti dal 58 al 68.Per la messa in onda si risponde soltanto ai punti 36,37,38,39,40,41.Per i ripetitori di programmi esteri o nazionali non va compilata ovviamente la scheda di messa in onda, ma va indicato al punto 64 per il solo primo impianto della della catena, la stazione straniera o rai ricevuta, gli impianti successivi vanno trattati normalmente.

Radio: AFN AMERICAN FORCES NETWORK 80
Denominazione Emittente

37 81
N° Impianto

128 83
Altezza centro sistema radiante dal

KATHREIN 84
Marca Pannelli o Antenne

DIPOLI 85
Tipo

X 86
Non Direttivo

6,58 87
Guadagno

V 88
Polarizzazione

Lombardi Ugo
American Forces Network
AFN Naples The Eagle
prot. 900552

89
Direttivo

90
Guad Direz Max

91
Direz Max Irradiazione

26,11 92
Pot (kW) Max Irradiazione

MINISTERO dello SVILUPPO ECONOMICO - Ispettorato Territoriale Campania (NA)

0 dBk	30 dBk	60 dBk	90 dBk	120 dBk	150 dBk	180 dBk	210 dBk	240 dBk	270 dBk	300 dB	330 dBk	
7.7	10.6	12.7	13.4	13.5	13.4	12.7	10.6	7.7	6.5	6.5	6.5	93

Angolo 1 2 [ ] 97

Angolo 4 1 [ ] 99

FACCIA 1

N tot Ant faccia 1: 4 95

Azimut faccia 1: 120 96

Pot totale sulla faccia 1 in kW: 3,97 98

Inclinazione elettrica o meccanica della faccia 1 in gradi: 4 100

FACCIA 2

N totale di pannelli sulla faccia 2 [ ] 101

Azimut faccia 2 [ ] 103

Pot totale sulla faccia 2 in kW [ ] 105

Inclinazione elettrica o meccanica della faccia 2 in gradi [ ] 107

FACCIA 4

N totale di pannelli sulla faccia 4 [ ] 102

Azimut faccia 4 [ ] 104

Pot tot faccia 4 [ ] 106

Inclinazione elettrica o meccanica della faccia 4 in gradi [ ] 108

FACCIA 3

Numero totale di pannelli sulla faccia 3 [ ] 109

Azimut faccia 3 [ ] 110

Pot totale sulla faccia 3 in kW [ ] 112

Inclinazione elettrica o meccanica della faccia 3 in gradi [ ] 114

Angolo 2 3 [ ] 111

Angolo 3 4 [ ] 113

APPENDIX A3 – COVERAGE MAPS

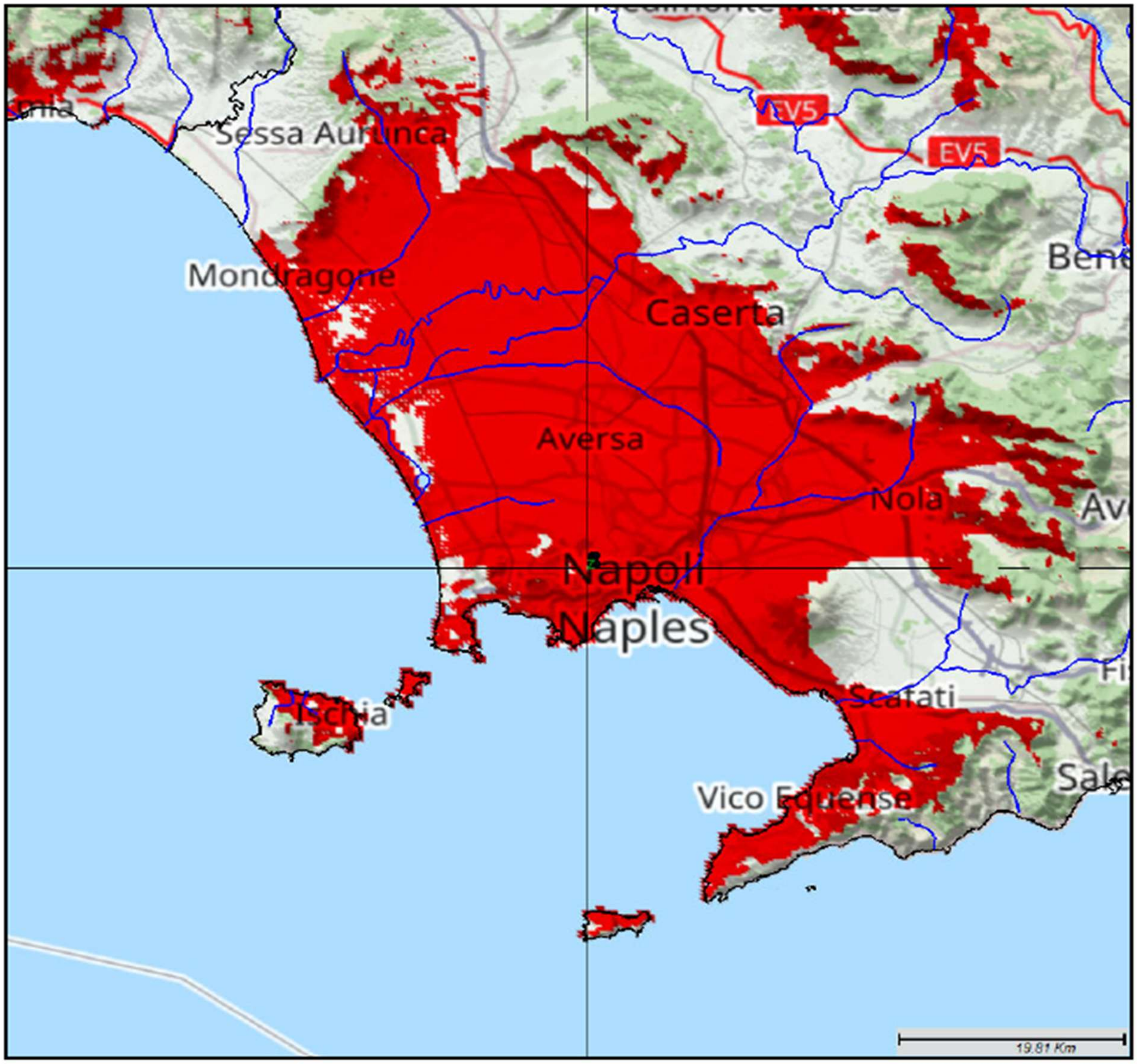
TX station: AFN Naples

Frequency: 97.30 MHz

Gain solid integration : enabled

Locality: Camaldoli 97.3 MHz

Coverage Area



dBµV/m	ppl		Km²	
≥ 74.0	2893294	77.5%	2295.1	15.9%
< 74.0	838450	22.5%	4663.2	32.4%

The physical phenomena included within the calculation of the propagation model 1812 are the following:
- Tropospherical scatter;
- Ducting and layer reflection/refraction;
- Terminal surroundings;
- Localtion variability;
- Delta Bullington + Spherical Diffraction

System of coordinates : WGS84

Longitude : 14°12'02.065", Latitude : 40°51'36.434"

Transmitter : 397.4mslm H128.0m

TX Power : 5000.00 W

Alpha 24, Ftdc, Fpr Power (Watt) : 5000.00W

Side : 120.0 Km Calculation resolution : 400 m

Boundary Hydrography

Propagation Model : ITU-R 1812-6

Time: 50% Locations: 50%

Receiver height (m) agl : 10.0

Area : Open

Powered by EmLAB - V3.26.1.1

Telecomunicazioni Aldena S.r.l.

AFN Naples

GOOGLE EARTH COVERAGE AREA
(signal strength > 74 dBuV/*m*) *

* as per ITU (International Telecommunication Union) recc. ITU-R 412-9 in the metropolitan areas